LESSON PLAN

COMPUTER (26)

Class	Air Agniveer Apprentice (Engineering)	**Lesson Plan No.**	05
Total No. of Lesson Plans	10	**Time Allotted for the Session**	80 min

PART A: GUIDELINES / GENERAL INSTRUCTIONS FOR INSTRUCTORS

1. Topic	Introduction to sonar system
2. Total No. of Sessions for this Topic	01 / 01
3. Sequence of this Session	01 / 01
4. Syllabus Extract	Introduction to sonar system. Purpose and technical details of active and passive sonar system.
5. Specific Objectives	26.1.1.1 Trainee will be able to understand the basic theory of communication.

6. List of Training Aids

Ser	Description	Remarks
(a)	BR/Text Book/Docket	Refer to the master syllabus reference list
(b)	Wall Charts	Nil
(c)	Films/Duration	Nil
(d)	PPT	Objective: teach the syllabus extract for this lesson
(e)	Models	Nil
(f)	CBT	Nil

7. Practical in Lab/Equipment Rooms/Simulators	Review of sonar theory diagrams; Safety: No live equipment handling; Supervision: Instructor monitors diagram interpretation.
8. Existing Knowledge of the Class	As per course prerequisites

PART B: DEVELOPMENT OF LESSON

9. Checking of Previous Knowledge / Introduction	Time: 10 min

(a) What is the primary function of a sonar system?
(b) Define the basic theory of communication in underwater contexts.
(c) List two key differences between active and passive sonar.

10. Development of Main Lesson

Sub	Content	Method	Trg Aids	Time
(a)	• Basic theory of underwater communication • Sound propagation in water medium • Definition and purpose of sonar systems • Fundamental components of sonar architecture	L & Ds	PPT, IB	15 min
(b)	• Technical details of active sonar operation • Technical details of passive sonar operation • Comparison of active and passive modes • Applications of active sonar in navigation • Applications of passive sonar in detection	L & Ds	PPT, PDF	20 min
(c)	• Limitations of active sonar systems • Limitations of passive sonar systems • Environmental factors affecting sonar performance • Basic troubleshooting of sonar signals • Integration of sonar with other sensors	Practical	IB, PDF	15 min

11. Summary and Evaluation

Method	Trg Aids	Time
Q & A	IB	15 min

(a) Explain the operational principle of an active sonar pulse.
(b) Describe how passive sonar detects targets without emitting sound.
(c) Identify three environmental factors that degrade sonar accuracy.

12. Home Assignment and Follow Up	Prepare a brief note on the transition from sonar theory to sonobuoy deployment principles for the next lesson.

Time	5 min

PART C: PREPARATION FOR TOMORROW (PFT) DIRECTIVES FOR TRAINEES

Ser	Syllabus for PFT	References	Related LPs
1	Introduction to sonobuoy system. Purpose and technical details of active and passive sonobuoy.	Refer to the master syllabus reference list	LP 06

PART D: PERSONAL EXPERIENCES (PE) OF INSTRUCTOR

Ser	Item	Remarks
(a)	Adequacy of time	
(b)	Suitability of training material/aids and additional aids required	
(c)	Doubts/intelligent questions raised by trainees	
(d)	Difficulties faced by instructor	
(e)	Other experience	

Instructor	**HoF (E/L)**
Signature: _____	Signature: _____

PART E: REVIEW SHEET

Date	Instructor's Comment	HOF (E/L)	Oi/C TDEC Remarks

Prepared by / Approved by

Prepared by	Approved by
Name, Rank, P No.	Name, Rank, P No.
Nov 26	Nov 26

Queries for Instructor

1. Confirm the master syllabus reference book/BR for this subject.
2. Confirm the class existing knowledge and prerequisites.
3. Confirm the session duration and sequence; 80 min and 01/01 were assumed.
4. Prepared-by/approved-by names, ranks, P numbers and dates must be filled before approval.
5. Confirm any training aids available beyond L & Ds, IB, PDF and PPT.

Timing Check

Introduction 10 + Main 50 + Summary 15 + Home 5 = 80 min